

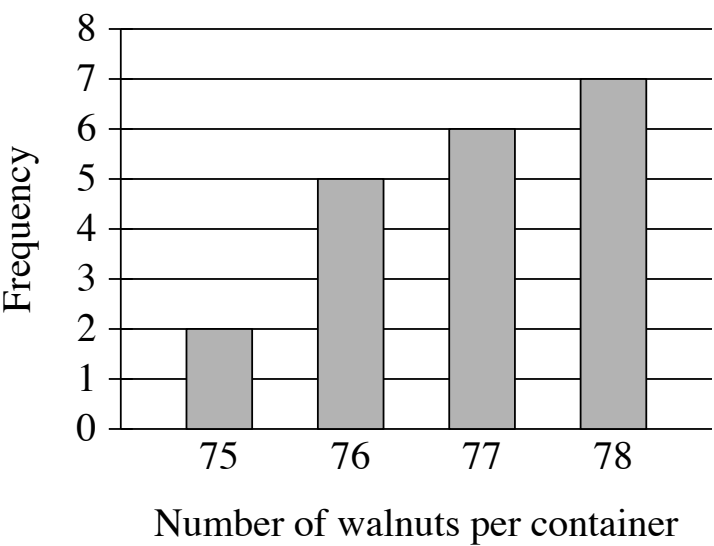
Question ID fe6a49d6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: fe6a49d6

PSAT2 M 2.1

The bar graph shows the distribution of the number of walnuts per container for **20** containers at a grocery store.



How many of these containers of walnuts contain exactly **78** walnuts?

- A. **2**
- B. **7**
- C. **20**
- D. **78**

ID: fe6a49d6 Answer

Correct Answer: B

Rationale

Choice B is correct. The height of each bar in the graph shown represents the number of containers that contain the number of walnuts specified at the bottom of the bar. The bar for 78 walnuts has a height of 7. Therefore, 7 of these containers of walnuts contain exactly 78 walnuts.

Choice A is incorrect. This is the number of containers that contain exactly 75 walnuts, not 78 walnuts.

Choice C is incorrect. This is the total number of containers of walnuts represented in the bar graph, not the number that contain exactly 78 walnuts.

Choice D is incorrect. This is the number of walnuts in a container that contains exactly 78 walnuts, not the number of containers that contain exactly 78 walnuts.

Question Difficulty: Easy

Question ID 771ee744

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 771ee744

PSAT2 M 2.2

There are **320** marbles in a container. Of these marbles, **10%** are red. How many marbles in the container are red?

- A. **32**
- B. **288**
- C. **320**
- D. **352**

ID: 771ee744 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that 10% of the 320 marbles in a container are red. Therefore, the number of red marbles can be calculated by multiplying the number of marbles in the container by $\frac{10}{100}$, which gives $320\frac{10}{100}$, or 32.

Choice B is incorrect. This is the number of marbles in the container that aren't red.

Choice C is incorrect. This is the total number of marbles in the container.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID fe4c1c9e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: fe4c1c9e

PSAT2 M 2.3

A mechanical device in a workshop produces items at a constant rate of **60** items per hour. At this rate, how many items will the mechanical device produce in **3** hours?

ID: fe4c1c9e Answer

Correct Answer: 180

Rationale

The correct answer is 180. It’s given that a mechanical device produces items at a constant rate of 60 items per hour. This rate can be written as $\frac{60 \text{ items}}{1 \text{ hour}}$. Let x represent the number of items the mechanical device will produce in 3 hours at the given rate. It follows that $\frac{60 \text{ items}}{1 \text{ hour}} = \frac{x \text{ items}}{3 \text{ hours}}$, which can be written as $\frac{60}{1} = \frac{x}{3}$, or $60 = \frac{x}{3}$. Multiplying each side of this equation by 3 yields $180 = x$. Therefore, at the given rate, the mechanical device will produce 180 items in 3 hours.

Alternate approach: It's given that a mechanical device produces items at a constant rate of 60 items per hour. At this rate, the mechanical device will produce $\frac{60 \text{ items}}{1 \text{ hour}} \cdot 3 \text{ hours}$, or 180 items in 3 hours.

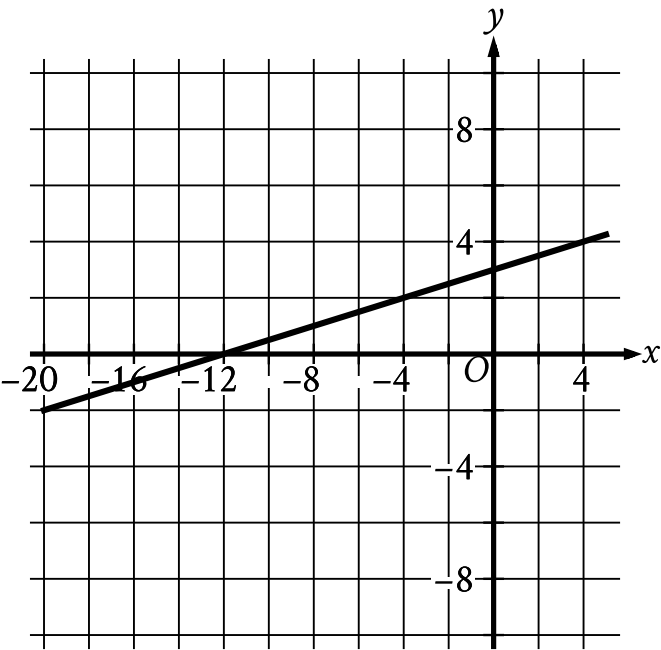
Question Difficulty: Easy

Question ID c10ad793

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: c10ad793

PSAT2 M 2.4



The graph of the linear function f is shown, where $y = f(x)$. What is the x-intercept of the graph of f ?

- A. $(-12, 0)$
- B. $(0, 0)$
- C. $(\frac{1}{4}, 0)$
- D. $(12, 0)$

ID: c10ad793 Answer

Correct Answer: A

Rationale

Choice A is correct. The x-intercept of a graph is the point where the graph intersects the x-axis. The graph of function f , where $y = f(x)$, intersects the x-axis at -12, 0. Therefore, the x-intercept of the graph of f is -12, 0.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID c6a26e14

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: c6a26e14

PSAT2 M 2.5

$|x + 45| = 48$ What is the positive solution to the given equation?

- A. 3
- B. 48
- C. 93
- D. 96

ID: c6a26e14 Answer

Correct Answer: A

Rationale

Choice A is correct. The given absolute value equation can be rewritten as two linear equations: $x + 45 = 48$ and $-x + 45 = 48$, or $x + 45 = -48$. Subtracting 45 from both sides of the equation $x + 45 = 48$ yields $x = 3$. Subtracting 45 from both sides of the equation $x + 45 = -48$ yields $x = -93$. Thus, the given equation has two possible solutions, 3 and -93. Therefore, the positive solution to the given equation is 3.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

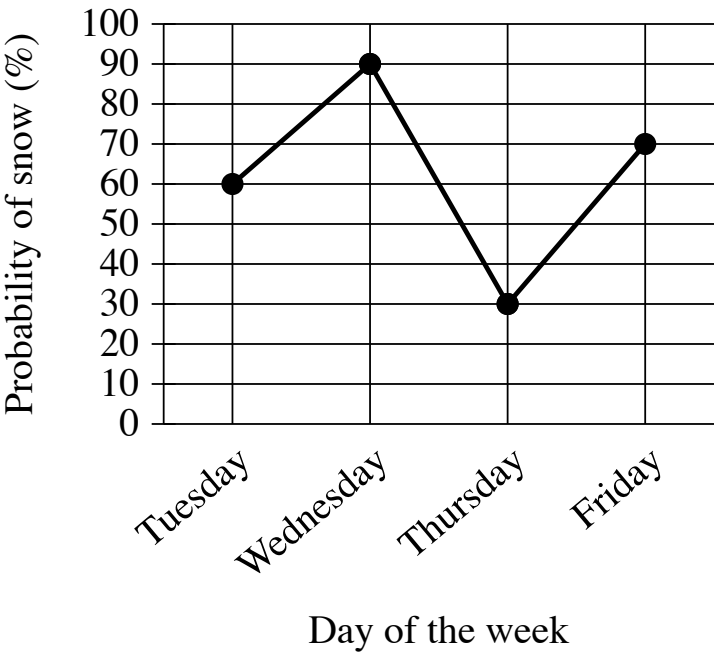
Question Difficulty: Easy

Question ID 13f67ddc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 13f67ddc

PSAT2 M 2.6



The line graph shows the probability of snow, as a percent, at a certain location for each day during a four-day period. According to the line graph, for which day during this four-day period is the probability of snow **30%**?

- A. Tuesday
- B. Wednesday
- C. Thursday
- D. Friday

ID: 13f67ddc Answer

Correct Answer: C

Rationale

Choice C is correct. For the line graph shown, the probability of snow, as a percent, is represented on the vertical axis. According to the line graph, during this four-day period, the probability of snow is 30% for Thursday.

Choice A is incorrect. The probability of snow on Tuesday is 60%.

Choice B is incorrect. The probability of snow on Wednesday is 90%.

Choice D is incorrect. The probability of snow on Friday is 70%.

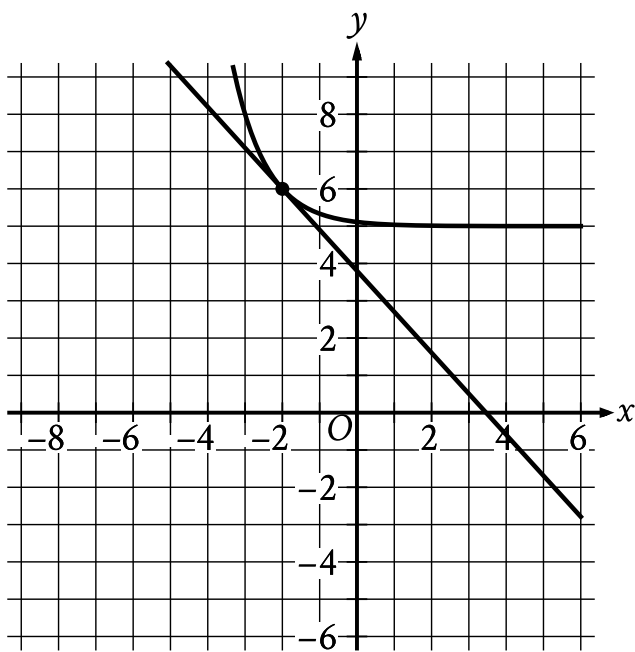
Question Difficulty: Easy

Question ID 5c7d5744

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 5c7d5744

PSAT2 M 2.7



The graph of a system of a linear equation and a nonlinear equation is shown. What is the solution (x, y) to this system?

- A. $(6, 0)$
- B. $(-2, 6)$
- C. $(0, -2)$
- D. $(0, 0)$

ID: 5c7d5744 Answer

Correct Answer: B

Rationale

Choice B is correct. The solution x, y to the system of two equations corresponds to the point where the graphs of the equations intersect in the xy -plane. The graphs of the linear equation and the nonlinear equation shown intersect at the point $-2, 6$. Thus, the solution x, y to this system is $-2, 6$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID e0177f5f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: e0177f5f

PSAT2 M 2.8

$x = 4$ $y = 5 - x$ The solution to the given system of equations is (x, y) . What is the value of y ?

- A. 1
- B. 4
- C. 5
- D. 9

ID: e0177f5f Answer

Correct Answer: A

Rationale

Choice A is correct. The first equation in the given system of equations is $x = 4$. Substituting 4 for x in the second equation in the given system of equations yields $y = 5 - 4$, or $y = 1$.

Choice B is incorrect. This is the value of x in the solution to the given system of equations, not the value of y .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID e006209c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: e006209c

PSAT2 M 2.9

A geologist needs to collect at least **67** samples of lava from a volcano. If the geologist has already collected **63** samples from the volcano, what is the minimum number of additional samples the geologist needs to collect?

- A. **130**
- B. **63**
- C. **4**
- D. **0**

ID: e006209c Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the geologist has already collected 63 samples from the volcano. Let x represent the number of additional samples the geologist needs to collect. After collecting x additional samples, the geologist will have collected a total of $63 + x$ samples. It's given that the geologist needs to collect at least 67 samples. Therefore, $63 + x \geq 67$. Subtracting 63 from each side of this inequality yields the inequality $x \geq 4$. Thus, the geologist needs to collect a minimum of 4 additional samples.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the number of samples the geologist has already collected, rather than the minimum number of additional samples the geologist needs to collect.

Choice D is incorrect. If the geologist collects 0 additional samples, the geologist will have collected a total of 63 samples, which is less than 67 samples.

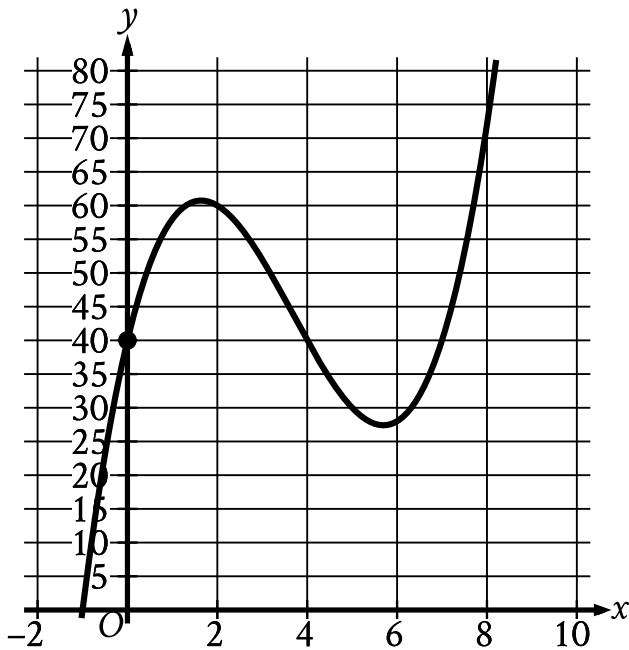
Question Difficulty: Easy

Question ID 26f5269a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 26f5269a

PSAT2 M 2.10



The y -intercept of the graph shown is (x, y) . What is the value of y ?

ID: 26f5269a Answer

Correct Answer: 40

Rationale

The correct answer is 40. The y -intercept of a graph in the xy -plane is the point x, y on the graph where $x = 0$. The y -intercept of the graph shown is $0, 40$. Therefore, the value of y is 40.

Question Difficulty: Easy

Question ID f2b63f49

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: f2b63f49

PSAT2 M 2.11

$8x - 7x + 130 = 260$ What value of x is the solution to the given equation?

ID: f2b63f49 Answer

Correct Answer: 130

Rationale

The correct answer is 130. It’s given that $8x - 7x + 130 = 260$. Combining like terms on the left-hand side of this equation yields $x + 130 = 260$. Subtracting 130 from each side of this equation yields $x = 130$. Therefore, the value of x that’s the solution to the given equation is 130.

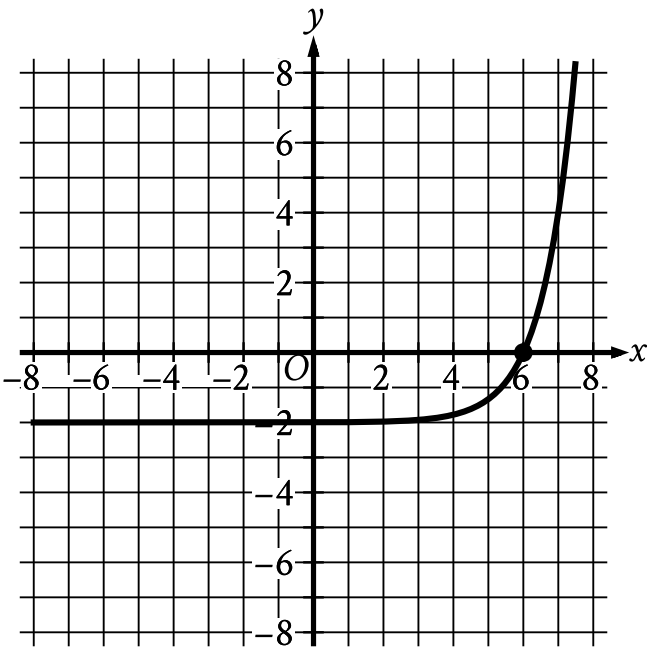
Question Difficulty: Easy

Question ID 2b6c12eb

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 2b6c12eb

PSAT2 M 2.12



What is the x-coordinate of the x-intercept of the graph shown?

ID: 2b6c12eb Answer

Correct Answer: 6

Rationale

The correct answer is 6. An x-intercept of a graph is a point on the graph where it intersects the x-axis, or where the value of y is 0. The graph shown intersects the x-axis at the point $(6, 0)$. Therefore, the x-coordinate of the x-intercept of the graph shown is 6.

Question Difficulty: Easy

Question ID 1d18794b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 1d18794b

PSAT2 M 2.13

A contract for a certain service requires a onetime activation cost of **\$35** and a monthly cost of **\$23**. Which equation represents this situation, where c is the total cost, in dollars, of this service contract for t months?

- A. $c = \frac{t}{23} + 35$
- B. $c = \frac{t}{35} + 23$
- C. $c = 23t + 35$
- D. $c = 35t + 23$

ID: 1d18794b Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that this service contract requires a monthly cost of \$ 23. A monthly cost of \$ 23 for t months results in a cost of \$ $23t$. It's also given that this service contract requires a onetime activation cost of \$ 35. Adding the onetime activation cost to the monthly cost of the service contract for t months yields the total cost c , in dollars, of this service contract for t months. Therefore, this situation can be represented by the equation $c = 23t + 35$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 4e97f862

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 4e97f862

PSAT2 M 2.14

The function f is defined by $f(x) = 3x - 8$. What is the value of $f(7)$?

- A. 29
- B. 13
- C. -5
- D. -29

ID: 4e97f862 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the function f is defined by $f(x) = 3x - 8$. The value of $f(7)$ is the value of $f(x)$ when $x = 7$. Substituting 7 for x in the given equation yields $f(7) = 3(7) - 8$, which is equivalent to $f(7) = 21 - 8$, or $f(7) = 13$.

Choice A is incorrect. This is the value of $f(7)$ when $f(x) = 3x + 8$, rather than $f(x) = 3x - 8$.

Choice C is incorrect. This is the value of $f(1)$, rather than $f(7)$. Choice D is incorrect. This is the value of $f(-7)$, rather than $f(7)$.

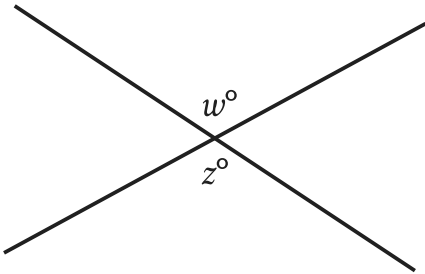
Question Difficulty: Easy

Question ID 64d1f49f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 64d1f49f

PSAT2 M 2.15



Note: Figure not drawn to scale.
In the figure, two lines intersect at a point. If $w = 136$, what is the value of z ?

- A. 36
- B. 44
- C. 68
- D. 136

ID: 64d1f49f Answer

Correct Answer: D

Rationale

Choice D is correct. In the figure shown, the angles with measures w° and z° are vertical angles. Since vertical angles are congruent, $w = z$. Therefore, if $w = 136$, the value of z is 136.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the measure, in degrees, of an angle that's supplementary, not congruent, to the angle with measure w° .

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 4efea6a3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 4efea6a3

PSAT2 M 2.16

The area of a rectangle is **57** square inches. The length of the longest side of the rectangle is **19** inches. What is the length, in inches, of the shortest side of this rectangle?

ID: 4efea6a3 Answer

Correct Answer: 3

Rationale

The correct answer is 3. The area of a rectangle can be calculated by multiplying the length of its longest side by the length of its shortest side. It’s given that the area of the rectangle is 57 square inches and the length of the longest side of the rectangle is 19 inches. Let x represent the length, in inches, of the shortest side of this rectangle. It follows that $57 = 19x$. Dividing both sides of this equation by 19 yields $3 = x$. Therefore, the length, in inches, of the shortest side of the rectangle is 3.

Question Difficulty: Easy

Question ID 02489d55

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	

ID: 02489d55

PSAT2 M 2.17

Which expression is equivalent to $19(x^2 - 7)$?

- A. $19x^2 - 133$
- B. $19x^2 - 26$
- C. $19x^2 - 7$
- D. $19x^2 + 12$

ID: 02489d55 Answer

Correct Answer: A

Rationale

Choice A is correct. The expression $19x^2 - 7$ can be rewritten as $19x^2 - 197$, which is equivalent to $19x^2 - 133$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID bd4d0e0c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: bd4d0e0c

PSAT2 M 2.18

The function f is defined by $f(x) = 10x^2 - 32x - 152$. What is the value of $f(0)$?

- A. -152
- B. -32
- C. 0
- D. 10

ID: bd4d0e0c Answer

Correct Answer: A

Rationale

Choice A is correct. The value of $f0$ is the value of fx when $x = 0$. The function f is defined by $fx = 10x^2 - 32x - 152$. Substituting 0 for x in this equation yields $f0 = 100^2 - 320 - 152$. This equation can be rewritten as $f0 = 100 - 0 - 152$, or $f0 = -152$. Therefore, the value of $f0$ is -152.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 0bebc08c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 0bebc08c

PSAT2 M 2.19

$x = 3$ $y = (15 - x)^2$ A solution to the given system of equations is (x, y) . What is the value of xy ?

- A. 432
- B. 54
- C. 45
- D. 18

ID: 0bebc08c Answer

Correct Answer: A

Rationale

Choice A is correct. The first equation in the given system of equations is $x = 3$. Substituting 3 for x in the second equation in the given system of equations yields $y = 15 - 3^2$, or $y = 144$. Substituting 3 for x and 144 for y in the expression xy yields $(3)(144)$, or 432. Therefore, the value of xy is 432.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 301faf80

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 301faf80

PSAT2 M 2.20

The product of two positive integers is **462**. If the first integer is **5** greater than twice the second integer, what is the smaller of the two integers?

ID: 301faf80 Answer

Correct Answer: 14

Rationale

The correct answer is 14. Let x represent the first integer and y represent the second integer. If the first integer is 5 greater than twice the second integer, then $x = 2y + 5$. It's given that the product of the two integers is 462; therefore $xy = 462$. Substituting $2y + 5$ for x in this equation yields $(2y + 5)(y) = 462$, which can be written as $2y^2 + 5y = 462$. Subtracting 462 from each side of this equation yields $2y^2 + 5y - 462 = 0$. The left-hand side of this equation can be factored by finding two values whose product is $2(-462)$, or -924, and whose sum is 5. The two values whose product is -924 and whose sum is 5 are 33 and -28. Thus, the equation $2y^2 + 5y - 462 = 0$ can be rewritten as $2y^2 - 28y + 33y - 462 = 0$, which is equivalent to $2y(y - 14) + 33(y - 14) = 0$, or $(2y + 33)(y - 14) = 0$. By the zero product property, it follows that $2y + 33 = 0$ or $y - 14 = 0$. Subtracting 33 from both sides of the equation $2y + 33 = 0$ yields $2y = -33$. Dividing both sides of this equation by 2 yields $y = -\frac{33}{2}$. Since y is a positive integer, the value of y isn't $-\frac{33}{2}$. Adding 14 to both sides of the equation $y - 14 = 0$ yields $y = 14$. Substituting 14 for y in the equation $xy = 462$ yields $x(14) = 462$. Dividing both sides of this equation by 14 yields $x = 33$. Therefore, the two integers are 14 and 33, so the smaller of the two integers is 14.

Question Difficulty: Hard

Question ID df747160

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	

ID: df747160

PSAT2 M 2.21

Which expression is equivalent to $17(x^2 - 100y^2)$?

- A. $17(x - 2y)(x - 50y)$
- B. $17(x - 2y)(x + 50y)$
- C. $17(x - 10y)(x - 10y)$
- D. $17(x - 10y)(x + 10y)$

ID: df747160 Answer

Correct Answer: D

Rationale

Choice D is correct. Expressions in the form $a^2 - b^2$ follow the difference of two squares pattern and can be factored as $(a - b)(a + b)$. In the given expression, $17(x^2 - 100y^2)$, the expression $x^2 - 100y^2$ follows the difference of two squares pattern. It follows that the expression $x^2 - 100y^2$ can be rewritten as $(x - 10y)(x + 10y)$. Therefore, the expression $17(x - 10y)(x + 10y)$ is equivalent to $17(x^2 - 100y^2)$.

Choice A is incorrect. This expression is equivalent to $17(x^2 - 52xy + 100y^2)$, not $17(x^2 - 100y^2)$.

Choice B is incorrect. This expression is equivalent to $17(x^2 + 48xy - 100y^2)$, not $17(x^2 - 100y^2)$.

Choice C is incorrect. This expression is equivalent to $17(x^2 - 20xy + 100y^2)$, not $17(x^2 - 100y^2)$.

Question Difficulty: Easy

Question ID 797a81fb

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	■ ■ ■

ID: 797a81fb

PSAT2 M 2.22

$-12x + 14y = 36$ $-6x + 7y = -18$ How many solutions does the given system of equations have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

ID: 797a81fb Answer

Correct Answer: D

Rationale

Choice D is correct. A system of two linear equations in two variables, x and y , has zero solutions if the lines representing the equations in the xy -plane are distinct and parallel. Two lines are distinct and parallel if they have the same slope but different y -intercepts. Each equation in the given system can be written in slope-intercept form $y = mx + b$, where m is the slope of the line representing the equation in the xy -plane and b is the y -intercept. Adding $12x$ to both sides of the first equation in the given system of equations, $-12x + 14y = 36$, yields $14y = 12x + 36$. Dividing both sides of this equation by 14 yields $y = \frac{6}{7}x + \frac{18}{7}$. It follows that the first equation in the given system of equations has a slope of $\frac{6}{7}$ and a y -intercept of $\frac{18}{7}$. Adding $6x$ to both sides of the second equation in the given system of equations, $-6x + 7y = -18$, yields $7y = 6x - 18$. Dividing both sides of this equation by 7 yields $y = \frac{6}{7}x - \frac{18}{7}$. It follows that the second equation in the given system of equations has a slope of $\frac{6}{7}$ and a y -intercept of $-\frac{18}{7}$. Since the slopes of these lines are the same and the y -intercepts are different, it follows that the given system of equations has zero solutions.

Alternate approach: To solve the system by elimination, multiplying the second equation in the given system of equations, $-6x + 7y = -18$, by -2 yields $12x - 14y = 36$. Adding this equation to the first equation in the given system of equations, $-12x + 14y = 36$, yields $(-12x + 12x) + (-14y + 14y) = 36 + 36$, or $0 = 72$. Since this equation isn't true, the given system of equations has zero solutions.

- Choice A is incorrect and may result from conceptual or calculation errors.
- Choice B is incorrect and may result from conceptual or calculation errors.
- Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard